

Nathan V. Whelan

Lab Address

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Research Interests

I am broadly interested in the evolution of biodiversity, how genetic diversity is shaped by changing environments, and the conservation of aquatic species. My research program seeks to understand historical relationships of organisms and how contemporary processes like anthropogenic climate change influence spatial genetic structure. My lab takes an interdisciplinary approach to studying biodiversity, conservation, and sports fish management.

Education

- 2013** Ph.D. Biological Sciences—The University of Alabama, Tuscaloosa, Alabama
Dissertation: Systematics, Life History, and Conservation of *Leptoxis* (Gastropoda: Cerithioidea: Pleuroceridae).
Adviser: Phillip M. Harris.
- 2008** B.S. Biology. Cum Laude. Chemistry Minor, Spanish Minor—Truman State University, Kirksville, Missouri.

Current Positions

- August 2025-present** *Director Museum of Biological Diversity & Associate Professor*
The Ohio State University-Columbus, Ohio
- Strategic leadership for one of the largest natural history collections in the United States.
 - Collections-based research.
 - Genomics, systematics, taxonomy, and conservation.
- November 2013-present** *Research Associate*
Smithsonian Institution, National Museum of Natural History-Washington, DC
- Collection-based research projects involving the systematics and genomics of freshwater mollusks, primarily Pleuroceridae, Semisulcospiridae, and Unionidae.

Previous Positions

- July 2016-July 2025** *Director Southeast Conservation Genetics Lab*
US Fish and Wildlife Service-Auburn, Alabama
- Principal investigator of government research lab.
 - I employed adaptive leadership and technical expertise to lead a diverse group of researchers.
 - Technical approaches for conservation research included genome sequencing, RADseq, GT-seq, environmental DNA (eDNA), and bioinformatics.
- March 2018-July 2025** *Assistant Research Professor*
Auburn University-Auburn, Alabama
- Dual appointment with above position.
 - Faculty member in the School of Fisheries, Aquaculture, and Aquatic Sciences in the College of Agriculture.
 - Educating and advising students and postdocs.

August 2023-October 2023 *Head of Staff – Regional Director’s Office, Alaska*

US Fish and Wildlife Service-Anchorage, Alaska

- Developmental Assignment for USFWS’s Advanced Leadership Development Program.
- Adviser to Regional Director and Deputy Regional Director.
- Served as Acting Deputy Regional Director for 3 weeks.
- Liaison between members of Regional Directorate Team and Regional Director’s Office.

February 2014-July 2016 *Postdoctoral Fellow*

Auburn University-Auburn, Alabama **Adviser:** Kenneth Halanych

- Researched the evolution of non-bilaterian metazoans and other marine invertebrates using genomic tools.
- Explored performance of amino acid substitution models with phylogenomic sized datasets in simulation and with empirical data.
- Instructor of record for graduate-level bioinformatics class.

May 2013-August 2013 *Mollusk Research Assistant*

May 2011-August 2011

Alabama Department of Conservation and Natural Resources-Marion, Alabama

- Designed and implemented novel captive propagation and conservation protocols for freshwater snails.
- Curated collections from state-wide mollusk surveys.

May 2012-August 2012 *Predoctoral Fellow* **Adviser:** Ellen Strong

Smithsonian Institution; National Museum of Natural History-Washington, DC

- Performed a research project on morphology and genetic diversity of imperiled freshwater gastropods.

Peer-Reviewed Articles (*Undergraduate author, **Graduate student author, ‡postdoc author, ^Dual first author)

59. Donohoo, S.A.**, **N.V. Whelan.** (2026). High quality genome for the threatened Louisiana Pearlshell *Margaritifera hembeli* (Conrad, 1838). *Journal of Molluscan Studies*. 92: eyag016. DOI: [10.1093/mollus/eyag016](https://doi.org/10.1093/mollus/eyag016)
58. **Whelan, N.V.**, A. Baldwin*, J.T. Garner, P.D. Johnson, E.E. Strong. (2026). Molecular data and museum collections reframe Pleuroceridae (Gastropoda: Cerithioidea) diversity and conservation priorities in an overlooked freshwater biodiversity hotspot. *Molecular Phylogenetics and Evolution*. 218: 108556. DOI: [10.1016/j.ympev.2026.108556](https://doi.org/10.1016/j.ympev.2026.108556)
57. Rotar, C.L.**, D.R. DeVries, R.A. Wright, L. Bilenker, M.Z. Billor, **N.V. Whelan.** (2026). Evaluating the effects of three Alabama River dams on movements and population connectivity of Freshwater Drum and White Crappie using otolith microchemistry and genetics. *Transactions of the American Fisheries Society*. 155: 138-159. DOI: [10.1093/tafafs/vnaf052](https://doi.org/10.1093/tafafs/vnaf052)
56. **Whelan, N.V.** (2025). “Integrative Phylogenomics” Integrates Sources of Error (eLetter response to Steenwyk, J.K, N.L. King. Integrative phylogenomics positions sponges at the root of the animal tree. *Science*. 390: 751-756). DOI: [10.1126/science.adw9456](https://doi.org/10.1126/science.adw9456)

55. Donohoo, S.A.** , P.D. Johnson, **N.V. Whelan**. (2025). Extreme endemism in temperate springs: evidence from narrow-range endemic snails. *Freshwater Biology*. 70: e70149. [DOI: 10.1111/fwb.70149](https://doi.org/10.1111/fwb.70149)
54. Perkins, M.A., K.L. DeVilbiss, **N.V. Whelan**, R.A. Hoch, B.K. Jones, J.W. Mays, H.K. Evans, S.B. Benfield. (2025). A new endemic freshwater mussel (Bivalvia: Unionidae) genus and species, *Ligodonta obscura*, from the Yadkin-Pee Dee River drainage in North Carolina, USA. *ZooKeys*. 1248: 245-265. [DOI: 10.3897/zookeys.1248.152952](https://doi.org/10.3897/zookeys.1248.152952)
53. **Whelan, N.V.**, M.L. Buntin, A.D. Wright**, P.D. Johnson. (2025). Population genomics reveals how Pleuroceridae (Gastropoda: Cerithioidea) species respond to habitat degradation. *American Malacological Bulletin*. 41: 45-54. [DOI: 10.4003/006.041.0104](https://doi.org/10.4003/006.041.0104)
52. Adcock, M.C.** , K.R. Moles, N.L. Garrison, S.A. Donohoo** , **N.V. Whelan**. (2024). Population genomics of the endangered freshwater mussel, *Arcidens wheeleri*, in the Little River, Arkansas. *Journal of Molluscan Studies*. 90: eyae029. [DOI: 10.1093/mollus/eyae029](https://doi.org/10.1093/mollus/eyae029)
51. Truong, T.N.** , **N.V. Whelan**, S.A. Bullard. (2024). Life cycle of *Leuceruthrus stephanocauda* (Faust, 1921) Womble and Bullard, 2022 (Digenea: Azygiidae), a new azygiid phylogeny, and combined evidence of low host specificity to its invertebrate and vertebrate hosts. *Comparative Parasitology*. 91: 5-18. [DOI: 10.1654/1525-2647-91.1.5](https://doi.org/10.1654/1525-2647-91.1.5)
50. Rezac, C.R., R.J. Ellwanger, S.A. Donohoo** , P.D. Hartfield, A.S. Ruppel, D.S. Ruppel, M.D. Wagner, **N.V. Whelan**. (2024). Surveys that prioritize site number over time per site will result in better gastropod status assessments: a case study on the rediscovery of Big Black Rocksnail. *Biodiversity and Conservation*. 33: 1811-1825. [DOI: 10.1007/s10531-024-02829-6](https://doi.org/10.1007/s10531-024-02829-6)
49. Kroll, J.D.** , J.M. Hill, P.R. Kingsley Smith, M.R. Kendrick, E.L. Gooding, C. Fuchs, **N.V. Whelan**, S.A. Bullard. (2024). First detection of white spot syndrome virus (WSSV) and infectious hypodermal and hematopoietic necrosis virus (IHHNV) from wild-caught giant tiger prawn, *Penaeus monodon* Fabricius, 1798 (Penaeoidea: Penaeidae) from the Gulf of Mexico and Northwestern Atlantic Ocean. *BiolInvasions Records*. 13: 121-140. [DOI: 10.3391/bir.2024.13.1.11](https://doi.org/10.3391/bir.2024.13.1.11)
48. Truong, T.N.** , **N.V. Whelan**, P.D. Johnson, M.L. Buntin, S.A. Bullard. (2024) Description, life cycle, and phylogenetics of *Proterometra wigglewomble* n. sp. (Digenea: Azygiidae) from the Cahaba River, Alabama, U.S.A. *Systematic Parasitology*. 101: 9. [DOI: 10.1007/s11230-023-10132-1](https://doi.org/10.1007/s11230-023-10132-1)
47. Gomes-dos-Santos. A, E. Froufe, J. Pfeiffer, N. Johnson, C. Smith, A.M. Machado, L.F.C. Castro, V. Tu Do, A. Hattori, N. Garrison, **N.V. Whelan**, I.N. Bolotov, I.V. Vikhrev, A.V. Kondakov, M. Ghamizi, V. Prié, A.E. Bogan, M. Lopes Lima. (2023). A novel assembly pipeline and functional annotations for targeted sequencing: a case study on the globally threatened Margaritiferidae (Bivalvia: Unionida). *Molecular Ecology Resources*. 23: 1403-1422. [DOI: 10.1111/1755-0998.13802](https://doi.org/10.1111/1755-0998.13802)
46. **Whelan, N.V.** E.E. Strong, N.S. Gladstone** , Jason W. Mays. (2023). Using genomics, morphometrics, and environmental niche modeling to test the validity of a narrow-range endemic snail. *ZooKeys*. 1158: 91-120. [DOI: 10.3897/zookeys.1158.94152](https://doi.org/10.3897/zookeys.1158.94152)
45. **Whelan, N.V.**, N.A. Johnson, A.S. Williams, M.A. Perkins, C.E. Beaver, J.W. Mays. (2023). Hidden in the hills: phylogeny of the freshwater mussel genus *Alasmidonta* (Bivalvia: Unionidae) and

description of a new species. *Zoological Journal of the Linnean Society*. 198: 650-676.
[DOI: 10.1093/zoolinnean/zlac106](https://doi.org/10.1093/zoolinnean/zlac106)

44. **Whelan, N.V.**, K.M. Halanych. (2023) Available data do not rule out Ctenophora as the sister group to all other Metazoa. *Nature Communications*. 14: 711. [DOI: 0.1038/s41467-023-36151-6](https://doi.org/10.1038/s41467-023-36151-6)
43. Strong, E.E., J.T. Garner, P.D. Johnson, **N.V. Whelan**. (2022) A systematic revision of the genus *Juga* from the freshwaters of the Pacific Northwest, USA (Cerithioidea, Semisulcospiridae). *European Journal of Taxonomy*. 848: 1-97. [DOI: 10.5852/ejt.2022.848.1993](https://doi.org/10.5852/ejt.2022.848.1993)
42. Gladstone, N.S.***, N.L. Garrison*, T. Lane, P.D. Johnson, J.T. Garner, **N.V. Whelan**. (2022) Population genomics reveal low differentiation and complex demographic histories in a highly fragmented and endangered freshwater mussel. *Aquatic Conservation: Marine and Freshwater Ecosystems*. 32: 1235-1248. [DOI: 10.1002/aqc.3849](https://doi.org/10.1002/aqc.3849)
41. **Whelan, N.V.**, P.D. Johnson, J.T. Garner, N.L. Garrison*, E.E. Strong. (2022) Prodigious polyphyly in Pleuroceridae (Gastropoda: Cerithioidea). *Bulletin of the Society of Systematic Biologists*. 1: 8419. [DOI: 10.18061/bssb.v1i2.8419](https://doi.org/10.18061/bssb.v1i2.8419)
40. Gladstone, N.S.***, M.L. Niemiller, B. Hutchins, B. Schwartz, A. Czaja, M.E. Slay, **N.V. Whelan**. (2022) Subterranean freshwater gastropod biodiversity and conservation in the United States and Canada. *Conservation Biology*. 36: e13722. [DOI: 10.1111/cobi.13722](https://doi.org/10.1111/cobi.13722)
39. Gladstone, N.S.***, **N.V. Whelan**. (2022) Pushing barcodes to their limits: phylogenetic placement of *Fontigens* Pilsbry, 1933 (Caenogastropoda: Littorinimorpha: Truncatelloidea) and elevation of Fontigentidae Taylor, 1966. *Journal of Molluscan Studies*. 88: eyab038. [DOI: 10.1093/mollus/eyab038](https://doi.org/10.1093/mollus/eyab038)
38. Gladstone, N.S.***, E.B. Pieper**, B.J. Dinkins, G.R. Dinkins, **N.V. Whelan**. (2021) The land snails and slugs of Tennessee, USA: taxonomic composition, distribution, and an evaluation of state-wide spatial and taxonomic survey bias. *American Malacological Bulletin*. 38: 51-62. [DOI: 10.4003/006.038.0211](https://doi.org/10.4003/006.038.0211)
37. Redak, C.A.***, A.S. Williams, J.T. Garner, K.M. Halanych, **N.V. Whelan**. (2021) Population genomics of the Rough Hornsnail, *Pleurocera foremani*, following river impoundment. *Journal of Heredity*. 112: 635-645. [DOI: 10.1093/jhered/esab065](https://doi.org/10.1093/jhered/esab065)
36. **Whelan, N.V.** (2021) Phenotypic plasticity and the endless forms of freshwater gastropod shells. *Freshwater Mollusk Biology and Conservation*. 24: 87-103, [DOI: 10.31931/fmbc-d-20-00015](https://doi.org/10.31931/fmbc-d-20-00015)
35. Garrison, N.L.**, P.D. Johnson, **N.V. Whelan**. (2021) Conservation genomics reveals low genetic diversity and multiple parentage in the threatened freshwater mussel, *Margaritifera hembeli*. *Conservation Genetics*. 22: 217-231. [DOI: 10.1007/s10592-020-01329-8](https://doi.org/10.1007/s10592-020-01329-8)
34. Dutton, H.R.***, L.H. Du Preez, O. Verneau, **N.V. Whelan**, S.A. Bullard. (2021) First record of a polystome from the from the alligator snapping turtle, *Macrochelys temminckii* (Cryptodira: Chelydridae) or Mississippi; with comments on “*Neopolystoma orbiculare* (Stunkard, 1916)” and its junior subjective synonyms. *Journal of Parasitology*. 107: 74-78. [DOI: 10.1645/20-96](https://doi.org/10.1645/20-96)

33. Whelan, N.V., E.E. Strong. (2021) Genetic diversity and historical demography in *Bellamya* gastropods from the Lake Victoria ecoregion reveal early and severe population collapse. *Molecular Ecology*. 30: 361-363. DOI: [10.1111/mec.15711](https://doi.org/10.1111/mec.15711)
32. Ksepka, S.P.** , B. Hickson, N.V. Whelan, S.A. Bullard. (2020) A new species of *Myxobolus* Bütschli, 1882 (Bivalvulida: Myxobolidae) infecting stratum spongiosum of the cryptic Sicklefin Redhorse, *Moxostoma* sp., (Cypriniformes: Catostomidae) from the Little Tennessee River, North Carolina, USA. *Folia Parasitologica*. 67: 030. DOI: [10.14411/fp.2020.030](https://doi.org/10.14411/fp.2020.030)
31. Wright, A.D.* , N.L. Garrison*, A.S. Williams, P.D. Johnson, N.V. Whelan. (2020) Range reduction of Oblong Rocksnail, *Leptoxis compacta*, shapes riverscape genetic patterns. *PeerJ*. 8: e9789. DOI: [10.7717/peerj.9789](https://doi.org/10.7717/peerj.9789)
30. Roop, H.J., N.V. Whelan, A.S. Williams, J. Page. (2020) First record of occurrence and genetic characterization of a population of Northern Snakehead *Channa argus* (Cantor 1842) in Georgia, USA. *BiolInvasions Records*. 9: 842-852. DOI: [10.3391/bir.2020.9.4.18](https://doi.org/10.3391/bir.2020.9.4.18)
29. Townsend, J.P.** , M.G. Tassia** , A. Damian-Serrano** , N.V. Whelan, K.M. Halanych, A.M. Sweeney. (2020) A mesopelagic ctenophore representing a new family, with notes on family-level taxonomy in Ctenophora: *Vampyroctena delmarvensis* gen. nov. sp. nov. (Vampyroctenidae, fam. nov.). *Marine Biodiversity*. 50: 34. DOI: [10.1007/s12526-020-01049-9](https://doi.org/10.1007/s12526-020-01049-9)
28. Ksepka, S.P.** , N.V. Whelan, C.M. Whipps, S.A. Bullard. (2020) A new species of *Thelohanellostoma* Kudo, 1933 (Myxozoa: Bivalvulida) infecting skeletal muscle of Blacktail Shiner, *Cyprinella venusta* Girard, 1856 (Cypriniformes: Cyprinidae) in the Chattahoochee River Basin, Georgia. *Journal of Parasitology*. 106: 350-359. DOI: [10.1645/19-162](https://doi.org/10.1645/19-162)
27. Bogantes, V.E.** , N.V. Whelan, K. Webster*, A.R. Mahon, K.M. Halanych (2020) Unrecognized diversity of a scale worm, *Polyeunoa laevis* (Annelida: Polynoidae), that feeds on soft coral. *Zoological Scripta*. 48: 236-249. DOI: [10.1111/zsc.12400](https://doi.org/10.1111/zsc.12400)
26. Ksepka, S.P.** , J.M. Rash, N.V. Whelan, S.A. Bullard. (2019) A new species of *Myxobolus* (Myxozoa: Bivalvulida) infecting the medulla oblongata and nerve cord of Brook Trout *Salvelinus fontinalis* in southern Appalachia (New River, North Carolina, USA). *Parasitology Research*. 118: 3241-3252. DOI: [10.1007/s00436-019-06472-x](https://doi.org/10.1007/s00436-019-06472-x)
25. Whelan, N.V., M.P. Galaska** , B.N. Siple** , J.M. Weber* , P. D. Johnson, K.M. Halanych, B.S. Helms. (2019) Riverscape genetic variation, migration patterns, and morphological variation of the threatened Round Rocksnail, *Leptoxis ampla*. *Molecular Ecology*. 28: 1593-1610. DOI: [10.1111/mec.15032](https://doi.org/10.1111/mec.15032)
24. Bullard, S.A., J.R. Roberts** , M.B. Warren** , H.R. Dutton** , N.V. Whelan, C.F. Ruiz** , T.R. Platt, V.V. Tkach, S.V. Brant, K.M. Halanych. (2019) Neotropical turtle blood flukes: two new genera and species from the Amazon River basin with a key to genera and comments on a marine-derived parasite lineage in South America. *Journal of Parasitology*. 105: 497-523. DOI: [10.1645/19-27](https://doi.org/10.1645/19-27)
23. Strong, E.E., N.V. Whelan. (2019) Assessing the diversity of western North American *Juga* (Semisulcospiridae, Gastropoda). *Molecular Phylogenetics and Evolution*. 136: 87-103. DOI: [10.1016/j.ympev.2019.04.009](https://doi.org/10.1016/j.ympev.2019.04.009)

22. Warren, M.B.** , H.R. Dutton** , **N.V. Whelan**, R.P.E. Yanong, S.A. Bullard. (2019) First record of a species of Mermithidae Braun, 1883 infecting a decapod, *Palaemon paludosus* (Palaemonidae). *Journal of Parasitology*. 105: 237-247. DOI: [10.1645/18-168](https://doi.org/10.1645/18-168)
21. Warren, M.B.** , C.F. Ruiz** , **N.V. Whelan**, D.C. Kritsky, S.A. Bullard. (2019) *Gymnura hemecus bulbosus* gen. et sp. nov. (Digenea: Aporocotylidae) infecting smooth butterfly rays, *Gymnura micrura* (Myliobatiformes: Gymnuridae) in the northern Gulf of Mexico, with a taxonomic key and further evidence for monophyly of chondrichthyan blood flukes. *Parasitology Research*. 118: 751-762. DOI: [10.1007/s00436-018-06202-9](https://doi.org/10.1007/s00436-018-06202-9)
20. **Whelan, N.V.**, K.M. Kocot, T.P. Moroz, K. Mukherjee, P. Williams, G. Paulay, L.L. Moroz, K.M. Halanych. (2017) Ctenophore relationships and their placement as the sister group to all other animals. *Nature Ecology and Evolution*. 1: 1737-1746. DOI: [10.1038/s41559-017-0331-3](https://doi.org/10.1038/s41559-017-0331-3)
19. **Whelan, N.V.**, P.D. Johnson, J.T. Garner, E.E. Strong. (2017) On the identity of *Leptoxis taeniata* – a misapplied name for the federally threatened Painted Rocksnail (Cerithioidea: Pleuroceridae). *Zookeys*. 697: 21-36. DOI: [10.3897/zookeys.697.14060](https://doi.org/10.3897/zookeys.697.14060)
18. Tassia, M.G.** , **N.V. Whelan**, K.M. Halanych. (2017) Toll-like receptor pathway evolution in deuterostomes. *Proceedings of the National Academy of Sciences*. 114: 7055–7060. DOI: [10.1073/pnas.1617722114](https://doi.org/10.1073/pnas.1617722114)
17. Costa-Paiva E.M.* , **N.V. Whelan**, D.S. Waits*, S. Santos, C.G. Schrago, K.M. Halanych. (2017) Discovery and evolution of novel hemerythrin genes in annelid worms. *BMC Evolutionary Biology*. 17: 85. DOI: [10.1186/s12862-017-0933-z](https://doi.org/10.1186/s12862-017-0933-z)
16. **Whelan, N.V.**, K.M. Halanych. (2017) Who let the CAT Out of the Bag? Accurately dealing with substitutional heterogeneity in phylogenomics. *Systematic Biology*. 66: 232-255. DOI: [10.1093/sysbio/syw084](https://doi.org/10.1093/sysbio/syw084)
15. Yuanning, L.** , K.M. Kocot, **N.V. Whelan**, S.R. Santos, D.S. Waits, D.J. Thornhill, K.M. Halanych. (2017) Phylogenomics of tubeworms (Siboglinidae, Annelida) and comparative performance of different reconstruction methods. *Zoologica Scripta*. 46: 200-213. DOI: [10.1111/zsc.12201](https://doi.org/10.1111/zsc.12201)
14. Halanych, K.M.^ , **N.V. Whelan**^, K.M. Kocot, A.B. Kohn, L.L. Moroz. (2016) Miscues misplace sponges. *Proceedings of the National Academy of Sciences*. 113: E946-E947. DOI: [10.1073/pnas.1525332113](https://doi.org/10.1073/pnas.1525332113)
13. **Whelan, N.V.**, E.E. Strong. (2016) Morphology, molecules and taxonomy: extreme incongruence in pleurocerids (Gastropoda, Cerithioidea, Pleuroceridae). *Zoologica Scripta*. 45: 62-87. DOI: [10.1111/zsc.12139](https://doi.org/10.1111/zsc.12139)
12. **Whelan, N.V.** (2016) Radular morphology of extinct pleurocerids (Gastropoda: Cerithioidea: Pleuroceridae). *American Malacological Bulletin*. 33: 221-226. DOI: [10.4003/006.033.0202](https://doi.org/10.4003/006.033.0202)
11. **Whelan, N.V.**, K.M. Kocot, K.M. Halanych. (2015) Employing phylogenomics to resolve the relationships among cnidarians, ctenophores, sponges, placozoans, and bilaterians. *Integrative and Comparative Biology*. 55: 1084-1095. DOI: [10.1093/icb/icv037](https://doi.org/10.1093/icb/icv037)

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9. **Whelan, N.V.**, P.D. Johnson, P.M. Harris. (2015) Life-history traits and shell morphology in the genus *Leptoxis* Rafinesque, 1819 (Gastropoda: Cerithioidea: Pleuroceridae). *Journal of Molluscan Studies*. 81: 85-95. [DOI: 10.1093/mollus/EYU058](https://doi.org/10.1093/mollus/EYU058)
8. **Whelan, N.V.**, K.M. Kocot, S.R. Santos, K.M. Halanych. (2014) Transcriptome sequencing of nemerteans reveals a diverse suite of toxin genes. *Genome Biology and Evolution*. 6: 3314-3325. [DOI: 10.1093/gbe/evu258](https://doi.org/10.1093/gbe/evu258)
7. **Whelan, N.V.**, E.E. Strong. (2014) Seasonal reproductive anatomy and sperm storage in pleurocerid gastropods (Cerithioidea: Pleuroceridae). *Canadian Journal of Zoology*. 92: 989-995. [DOI: 10.1139/cjz-2014-0165](https://doi.org/10.1139/cjz-2014-0165)
6. Benstead, J.P., J.M. Hood, **N.V. Whelan**, M.R. Kendrick, D. Nelson, A.F. Hanninen, L.M. Demi. (2014) Dietary P-growth coupling across diverse fish taxa: a meta-analysis of experimental aquaculture studies. *Ecology*. 95: 2768-2777. [DOI: 10.1890/13-1859.1](https://doi.org/10.1890/13-1859.1)
5. Johnson, P.D., A.E. Bogan, K.M. Brown, N.M. Burkhead, J.R. Cordeiro, J.T. Garner, P.D. Hartfield, D.A. Lepitzki, G.L. Mackie, E. Pip, T.A. Tarpley, J.S. Tiemann, **N.V. Whelan**, E.E. Strong. (2013) Conservation status of freshwater gastropods of Canada and the United States. *Fisheries*. 38: 247-282. [DOI: 10.1080/03632415.2013.785396](https://doi.org/10.1080/03632415.2013.785396)
4. **Whelan N.V.**, P.D. Johnson, P.M. Harris. (2012) Rediscovery of *Leptoxis compacta* (Anthony, 1854) (Gastropoda: Cerithioidea: Pleuroceridae). *PLoS One*. 7: e42499. [DOI: 10.1371/journal.pone.0042499](https://doi.org/10.1371/journal.pone.0042499)
3. **Whelan, N.V.**, P.D. Johnson, P.M. Harris. (2012) Presence or absence of carinae between closely related populations of *Leptoxis ampla* (Anthony, 1854) (Gastropoda: Pleuroceridae) is not the result of ecophenotypic plasticity. *Journal of Molluscan Studies*. 78: 231-233. [DOI: 10.1093/mollus/eyu005](https://doi.org/10.1093/mollus/eyu005)
2. **Whelan, N.V.** (2011) Species tree inference in the age of genomics. *Trends in Evolutionary Biology*. 3: e5. [DOI: 10.4081/eb.2011.e5](https://doi.org/10.4081/eb.2011.e5)
1. **Whelan, N.V.**, A. Geneva, D.L. Graf. (2011) Molecular phylogenetic analysis of tropical freshwater mussels (Mollusca: Bivalvia: Unionoida) resolves the position of *Coelatura* and supports a monophyletic Unionidae. *Molecular Phylogenetics and Evolution*. 61: 504-514. [DOI: 10.1016/j.ympev.2011.07.016](https://doi.org/10.1016/j.ympev.2011.07.016)

Articles in Review or Revision (Graduate student author)**

1. Steed, A.E. **, **N.V. Whelan**. Conservation genomics of the Smooth Rocksnail, *Leptoxis virgata* (Gastropoda: Cerithioidea: Pleuroceridae). In Review at Conservation Genetics.
2. Hein, S.R., D.A. Trujillo, **N.V. Whelan**, C.R. Randklev, D.J. Berg. Do neutral and adaptive variation provide similar conservation insights? A case study of the federally endangered freshwater mussel, *Popenaias popeii*. In Revision for resubmission.

Popular press, non-peer reviewed reports, newsletter articles, and blogs (Graduate student author)**

28. **Whelan, N.V.**, A.E. Steed**, S.A. Donohoo**, A.P. Hannah. (2025) 2025 Report on Genotyping of Black Bass from Virginia. Submitted to Virginia Department of Wildlife Resources.
27. **Whelan, N.V.**, A.E. Steed**, S.A. Donohoo**, A.P. Hannah. (2025) 2025 Report on Genotyping of Black Bass from North Carolina. Submitted to North Carolina Department of Wildlife Resources.
27. **Whelan, N.V.**, J. Von Bargen, M.C. Adcock**, S.A. Donohoo**, A.P. Hannah. (2024) Report on Genotyping of Black Bass from Virginia. Submitted to Virginia Department of Wildlife Resources.
26. **Whelan, N.V.**, J. Von Bargen, M.C. Adcock**, S.A. Donohoo**, A.P. Hannah. (2024) Report on Genotyping of Black Bass from North Carolina. Submitted to North Carolina Department of Wildlife Resources.
25. **Whelan, N.V.**, A.P. Hannah. (2024) Report on the range-wide population genetics of Spotfin Chub (*Erimonax monachus*). Submitted to U.S. Fish and Wildlife Service Asheville Ecological Services Field Office.
24. **Whelan, N.V.**, J. VonBargen, M.C. Adcock**, S.A. Donohoo**, A.P. Hannah. (2023) Report on Genotyping of Black Bass from Kentucky. Submitted to Kentucky Department of Fish and Wildlife Resources.
23. Donohoo, S.A.** , **N.V. Whelan**. (2022) Report on species identify of putative *E. cochliaris* and *E. bellacrenata* (Gastropoda: Cerithioidea: Pleuroceridae) populations in central Alabama. Report submitted to USFWS Southeast Regional Office.
22. Gladstone, N.S.** , **N.V. Whelan**. (2021) Protecting the living jewels hidden underground: conservation needs of groundwater-restricted snails. *Current Conservation*. April 2021. Available at <https://www.currentconservation.org/protecting-the-living-jewels-hidden-underground-conservation-needs-of-groundwater-restricted-snails/>
21. **Whelan, N.V.** (2021) Genomics research reveals secrets of Rough Hornsnail. *FAC Outdoors*. November 2021.
20. Williams, A.S., **N.V. Whelan**. (2021) Molecular confirmation of Gopher Frog (*Lithobates capito*) eggs from South Carolina. Report submitted to USFWS and Bears Bluff National Fish Hatchery.
19. **Whelan, N.V.**, N.S. Gladstone**, A.S. Williams. (2021) Initial phylogenetic analyses support *Patera clarki nantahala* (Helicoidea: Polygyridae) as a valid subspecies. Report submitted to USFWS and Asheville Ecological Services Offices.
18. Gladstone, N.S.** , **N.V. Whelan**. (2021) Assessment of genetic diversity, effective population size, and population connectivity of the threatened Painted Rocksnail (*Leptoxis coosaensis*) and Lacy Elimia (*Elimia crenatella*) in the Mobile River Basin, Alabama.
17. Williams, A.S., **N.V. Whelan**. (2020) Assessment of hatchery contribution and genetic diversity of American shad (*Alosa sapidissima*) in the Edisto River, SC, an update. Report submitted to USFWS and Bears Bluff National Fish Hatchery.

16. **Whelan, N.V.**, N.L. Garrison, J.T. Garner, P.D. Johnson. (2020) Conservation genetics of listed and candidate freshwater snails in the Mobile and Tennessee River drainages, final report. Report submitted to Alabama Department of Conservation and Natural Resources.
15. Williams, A.S., **N.V. Whelan**. (2019) Molecular identification of Gopher Frog (*Rana capito*) raised at Bears Bluff National Fish Hatchery. Report submitted to USFWS and Bears Bluff National Fish Hatchery.
14. **Whelan, N.V.**, N.L. Garrison, J.T. Garner, P.D. Johnson. (2019) Conservation genetics of listed and candidate freshwater snails in the Mobile and Tennessee River drainages, interim report. Report submitted to Alabama Department of Conservation and Natural Resources.
13. Williams, A.S., **N.V. Whelan**. (2019) Assessment of hatchery contribution and genetic diversity of American shad (*Alosa sapidissima*) in the Edisto River, SC, an update. Report submitted to USFWS and Bears Bluff National Fish Hatchery.
12. Williams, A.S., **N.V. Whelan**. (2019) Genetic evaluation of Lake Sturgeon reintroduction program in the Tennessee River, 2019 Annual Report. Report submitted to USFWS and Tennessee River Lake Sturgeon Reintroduction Working Group.
11. Williams, A.S., **N.V. Whelan**. (2018) Assessment of hatchery contribution and genetic diversity of American shad (*Alosa sapidissima*) in the Edisto River, SC. Report submitted to USFWS and Bears Bluff National Fish Hatchery
10. Williams, A.S., **N.V. Whelan**. (2018) Genetic assessment of Striped Bass (*Morone saxatilis*) brood stock repository lakes in the Apalachicola-Chattahoochee-Flint (ACF) River Systems. Report submitted to USFWS and Gulf Strain Striped Bass Recovery Committee.
9. Williams, A.S., **N.V. Whelan**. (2018) Genetic evaluation of Lake Sturgeon reintroduction program in the Tennessee River, 2018 Annual Report. Report submitted to USFWS and Tennessee River Lake Sturgeon Reintroduction Working Group.
8. Helms, B.S., **N.V. Whelan**, L. Tolley-Jordan, K.M. Halanych, B.N. Siple, D. Wicker, J. Weber, M.P. Galaska. (2017) Population structure of the Round Rocksnail (*Leptoxis ampla*) in the Cahaba River. Report submitted to Alabama Department of Conservation and Natural Resources.
7. Strong, E.E., **N.V. Whelan**, J.T. Garner, P.D. Johnson. (2015) Genetic diversity of the Black Mudalia, *Elimia melanoides* (Conrad 1834) (Caenogastropoda, Pleuroceridae). Report submitted to U.S. Fish and Wildlife Service.
6. **Whelan, N.V.** (2014) Classical methods shed new light on freshwater snail reproduction, with conservation implications. *No Bones About It: NMNH's Invertebrate Biology Blog*. http://nmnh.typepad.com/no_bones/2014/11/classical-methods-shed-new-light-on-freshwater-snail-reproduction-with-conservation-implications.html
5. Evans, R., **N.V. Whelan**. (2014) Recent observation of *Leptoxis* from the Rockcastle River, Kentucky. *Ellipsaria*. 16: 4.
4. **Whelan, N.V.** (2014) Contemporary surveys and new explorations. *Cracking the Collections*. <https://crackingthecollections.wordpress.com/2014/07/10/new-explorations/>

3. **Whelan, N.V.** (2012) *Leptoxis compacta* (Gastropoda: Pleuroceridae) found for the first time in over 75 years. *Ellipsaria* (cover article). 14: 3.
2. **Whelan, N.V.** (2010) Preliminary results of life history strategies of the freshwater snail genus *Leptoxis* (Cerithioidea: Pleuroceridae) from the southeastern United States. *American Malacological Society Newsletter*. 41: 2.
1. **Whelan, N.V., P.M. Harris, P.D. Johnson.** (2010) Microsatellite DNA loci primers for *Leptoxis ampla* (Gastropoda: Pleuroceridae). Report submitted to Alabama Department of Conservation and Natural Resources.

Grants and Fellowships Over \$8,000,000 awarded since 2008. Note that listed funding from U.S. Fish and Wildlife Service are cooperative agreements with my past appointment at Auburn University and is additional funding that I have received that was not a part of my lab's base funding from congressional appropriations.

- 2026 **Nationwide Species Recovery Grant, National Fish and Wildlife Foundation.** *Conservation Genomics of Royal Marstonia* (Gastropoda: Hydrobiidae: *Marstonia*). (PI). \$64,718.
- 2024 **Cooperative Agreement, U.S. Fish and Wildlife Service.** *Invasive Species Early Detection Rapid Response Molecular Lab Network*. (PI). \$1,625,074.
- 2024 **Cooperative Agreement, U.S. Fish and Wildlife Service.** *Enhancing Infrastructure for Fish Disease Diagnostics and Environmental DNA*. (Co-PI). \$4,000,000.
- 2024 **Ecological Services Species Recovery Funding, U.S. Fish and Wildlife Service.** *Conservation Genomics of Frosted Flatwoods Salamander (Ambystoma cingulatum) on St. Marks National Wildlife Refuge*. (PI). \$50,000.
- 2024 **Contract for Black Bass Genotyping, North Carolina Wildlife Resources Convention.** *GT-Seq genotyping to assess Black Bass introgression and species ID from North Carolina*. (PI). \$20,898.
- 2024 **Contract for Black Bass Genotyping, Virginia Department of Wildlife Resources.** *GT-Seq genotyping to assess Black Bass introgression and species ID from Virginia*. (PI). \$30,240.
- 2023 **Contract for Black Bass Genotyping, Virginia Department of Wildlife Resources.** *GT-Seq genotyping to assess Black Bass introgression and species ID from Virginia*. (PI). \$36,800.
- 2023 **Contract for Black Bass Genotyping, Kentucky Department of Fish and Wildlife Resources.** *GT-Seq genotyping to assess Black Bass introgression and species ID from Kentucky*. (PI). \$31,552.
- 2023 **Contract for Black Bass Genotyping, North Carolina Wildlife Resources Convention.** *GT-Seq genotyping to assess Black Bass introgression and species ID from North Carolina*. (PI). \$50,747.
- 2023 **Supplemental Research Funding, U.S. Fish and Wildlife Service.** *Development of a Trispart Darter Genetics Management Plan*. (PI). \$35,000.
- 2021 **Competitive State Wildlife Grant, U.S. Fish and Wildlife Service.** *Advancing Conservation and Restoration of Brook Floater and Associated Freshwater Mussels*. (Co-PI/Auburn University component PI). \$1,332,306.43 (Auburn University Component for Genetics: \$96,112.29).
- 2020 **Competitive 1311 funds for Fisheries and Aquatic Conservation, U.S. Fish and Wildlife Service.** *Infrastructure and Standard Operating Procedure for Genetics Data Storage*. (PI). \$30,356.
- 2020 **Competitive 1311 funds for Fisheries and Aquatic Conservation, U.S. Fish and Wildlife Service.** *Genetic Technology Sharing Among Regions*. (PI). \$45,417.

- 2019 Sea Grant, National Oceanic and Atmospheric Administration.** *Pelagic Ecosystem Research Consortium (PERC) for highly migratory species (HMS) in the Gulf of Mexico and NW Atlantic Ocean.* (Co-PI). \$1,600,000 (Auburn University Component: \$400,000).
- 2019 Competitive 1311 funds for Fisheries and Aquatic Conservation, U.S. Fish and Wildlife Service.** *Systematics and phylogenetics of Myxobolus: a needed tool for preventing invasion and controlling spread of disease-causing fish parasites.* (PI). \$102,000.
- 2019 Competitive 1311 funds for Fisheries and Aquatic Conservation, U.S. Fish and Wildlife Service.** *Whole nuclear genome sequencing of a freshwater mussel and two fish.* (PI) \$59,000
- 2018 Aquatic Nuisance Species Grant, Gulf States Marine Fisheries Commission.** *Novel tools to detect, track, and trace Myxobolus cerebralis, (causative agent of whirling disease) in the southeastern United States.* (Co-PI). \$25,000.
- 2018 Endangered Species Research Funding, Ecological Services, U.S. Fish and Wildlife Service.** *Conservation genomics of Spotfin Chub Cyprinella monacha (Cypriniformes: Cyprinidae: Leuciscinae).* (PI) \$20,000.
- 2017 Competitive 1311 funds for Fisheries and Aquatic Conservation, U.S. Fish and Wildlife Service.** *Development of genetics management plans for six federally listed animal species.* (PI) \$372,151.
- 2017 Alabama Department of Conservation Section 6 Grant.** *Conservation genetics of listed and candidate freshwater snails in the Mobile and Tennessee River drainages.* (PI) \$62,067
- 2016 Smithsonian Institution Global Genome Initiative Research Grant.** *Targeted sequencing and phylogenomics of the critically imperiled Pleuroceridae.* (Co-PI) \$29,973
- 2016 Alabama Department of Conservation Section 6 Grant.** *Population structure of the Round Rocksnail (Leptoxis ampla) in the Cahaba River.* (PI) \$33,244
- 2014 Alabama Department of Conservation Section 6 Grant.** *Development of Nuclear Molecular Markers for Pleuroceridae Systematics and Conservation Assessment.* (Co-PI) \$21,143
- 2013 The University of Alabama Graduate Student Travel Award.** \$800
- 2012 Smithsonian Institution Predoctoral Fellowship.** *Do extreme genetic differences translate to internal morphological differences within "species" of Pleuroceridae (Gastropoda: Cerithioidea)?* \$7,650
- 2012 American Malacological Society Student Travel Award.** \$770
- 2012 The University of Alabama Graduate Student Travel Award.** \$1150
- 2011 National Science Foundation Doctoral Dissertation Improvement Grant.** *Remnants of a diverse past: assessing the phylogenetic position of recently extinct pleurocerid snails (Gastropoda: Pleuroceridae).* DEB-1110638. \$14,996
- 2011 The University of Alabama Graduate Council Fellowship.** \$36,356
- 2011 The University of Alabama Graduate School Student Research Grant.** \$600
- 2010 Alabama Department of Conservation and Natural Resources.** *Microsatellite Development Grant.* (Listed as subcontractor) \$5,000
- 2010 The University of Alabama Graduate Student Travel Award and International Travel Supplement.** \$900
- 2010 Conchologists of America Student Research Award.** \$1,400
- 2010 The University of Alabama Graduate Student Association Travel Grant.** \$400
- 2010 The University of Alabama Graduate School Travel Grant.** \$600

- 2009** **Birmingham Audubon Society Walter F. Coxe Research Grant.** \$1,000
- 2009** **American Malacological Society Melbourne R. Carriker Student Research Award.** \$800
- 2008** **The University of Alabama, Ecology, Evolution and Systematics 2-year enhancement fellowship.** \$55,000

Teaching Experience

- 2026** *Instructor.* EEOB 8896, Adaptive Leadership and Management. The Ohio State University.
- 2026** *Instructor.* EEOB 4410, Conservation Biology. The Ohio State University
- 2025** *Guest Lecturer.* Conservation Ecology of Freshwater Invertebrates. Auburn University.
- 2023** *Guest Lecturer.* Conservation Ecology of Freshwater Invertebrates. Auburn University.
- 2022** *Instructor.* Bioinformatics for Conservation Genomics week-long workshop. Auburn University
- 2021** *Guest Lecturer.* Limnology. Auburn University
- 2021** *Guest Lecturer.* Conservation Ecology of Freshwater Invertebrates. Auburn University.
- 2021** *Guest Lecturer.* Principles of Systematics. The University of Alabama.
- 2019** *Instructor.* Bioinformatics and Genome Assembly for Conservation Genetics week-long workshop. Auburn University.
- 2019** *Guest Lecturer.* Agriculture Genetics. Auburn University.
- 2019** *Guest Lecturer.* Conservation Ecology of Freshwater Invertebrates. Auburn University.
- 2018** *Instructor.* Gastropod Classification and Identification Workshop. Freshwater Mollusk Conservation Society.
- 2018** *Instructor.* Phylogenetics Crash Course Workshop. Auburn University.
- 2017** *Guest Lecturer.* Conservation Biology Learning Community. Auburn University.
- 2017** *Guest Lecturer.* Agriculture Genetics. Auburn University.
- 2017** *Guest Lecturer.* Conservation Ecology of Freshwater Invertebrates. Auburn University.
- 2016** *Guest Lecturer.* Conservation Biology Learning Community. Auburn University.
- 2016** *Guest Lecturer.* Agriculture Genetics. Auburn University.
- 2016** *Guest Lecturer.* Introduction to Bioinformatics. Auburn University.
- 2014** *Instructor.* Python for Bioinformatics. Auburn University.
- 2014** *Guest Lecturer.* Phylogenetics. Auburn University.
- 2014** *Guest Lecturer.* Invertebrate Biology. Auburn University.
- 2013** *Graduate Teaching Assistant.* Anatomy and Physiology II Lab. The University of Alabama.
- 2013** *Guest Lecturer.* Conservation Biology. The University of Alabama.
- 2012** *Graduate Teaching Assistant.* Vertebrate Zoology. The University of Alabama.
- 2012** *Guest Lecturer.* Principles of Systematics. The University of Alabama.
- 2012** *Guest Lecturer.* Genomics. The University of Alabama.
- 2011** *Graduate Teaching Assistant.* Introductory Biology II Lab. The University of Alabama.
- 2007-2008** *Undergraduate Teaching Assistant.* Chemistry for Non-Majors I & II. Truman State University.

- 2006** *Undergraduate Teaching Assistant. Introductory Biology II. Truman State University.*
- 2005** *Undergraduate Teaching Assistant. Introductory Biology I. Truman State University.*

Student and Postdoctoral Mentoring

2026 - present Conservation genomics and host evolution of freshwater mussels

I am the adviser and major professor for a PhD Student at The Ohio State University, Franchesca Ruiz, who is studying conservation genomics of freshwater mussels and examining patterns of host fish evolution in freshwater mussels.

2026 - present Biogeography of freshwater snails

I am the adviser and major professor for a masters student at The Ohio State University, Payton Carroll, who is studying biogeography and evolution of freshwater snails in the family Pleuroceridae

2024 - present Conservation genomics of freshwater mussels and snails

I co-advise a PhD student at Auburn University, Alicia Krause, who is studying the population genomics of the Alabama Lampmussel and Spider Elimia. Alicia is also performing eDNA surveys of fisheries pond at the Auburn University E.W. Shell Fisheries Center.

2024 - present Conservation genomics and systematics of Pleuroceridae gastropods

I co-advise a PhD student at Auburn University, Elizabeth Strasko, who is studying conservation genomics of the Painted Rocksnail and systematics of Mobile River drainage snails in the genus *Elimia*.

2024 - 2026 Conservation genomics of Seep Mudalia snail and Frosted Flatwoods Salamander

I was the adviser and major professor for a masters student at Auburn University, Annamarie Steed, who is studying conservation genomics of two critically imperiled aquatic species in the southeastern United States. Anna graduated with an MS in May 2026 and accepted a PhD position at Michigan State University.

2022 - 2026 Systematics of freshwater gastropods

I was the adviser and major professor for a PhD student at Auburn University, Sam Donohoo, who is studying the systematics of freshwater gastropods. Under my supervision, Sam received an Auburn University Presidential Fellowship. Sam graduated with a PhD in May 2026 and took a position as Curator of Zoology at the Sternberg Museum.

2022 - 2024 Conservation Genomics of Brook Floater and Ouachita Rock Pocketbook

I was the adviser and major professor for a masters student at Auburn University, Mia Adcock, who studied the conservation genomics of two critically imperiled freshwater mussels. Mia graduated with an MS in August 2024.

2022 U.S Fish and Wildlife Service Directorate Fellows Program

I was the project supervisor and research adviser for an undergraduate research, Raquel Gonzalez, who was a 2022 USFWS Directorate Fellow. This student led an independent research project on Alabama Pearlshell Mussel conservation genomics, including writing an official USFWS genetic management plan. Although the fellowship is over, I continue to mentor Raquel as she writes her senior thesis on the research done in my lab. The

student will be presenting their research at SICB 2023 and is working on a manuscript for submission to a peer-reviewed journal.

2022 NSF Research Experience for Undergraduates

I advised an undergraduate researcher, Annika Baldwin, as a part of the NSF funded REU warm-water aquatic ecology site at Auburn University. The student led an independent research project focused on conservation genomics of the critically imperiled freshwater gastropod *Elimia melanoides*. The student will be presenting their research at SICB 2023 and is working on a manuscript for submission to a peer-reviewed journal.

2020 - 2022 Systematics and conservation of Pleuroceridae

I advised a masters student, Nicholas Gladstone, who is working on mollusk conservation and landscape genomics. As a student in my lab, Nick received three student research grants from different scientific organizations and a best talk award from the American Malacological Society. Nick also published multiple papers while in my lab and graduated with a masters in 2022.

2019 - 2021 Conservation genomics of freshwater gastropods.

I co-advised a masters student, Caitlin Redak, on a conservation genetics project for the federally endangered freshwater gastropod *Pleurocera foremani*.

2019 NSF Research Experience for Undergraduates

I advised an undergraduate researcher, Aaliyah Wright, as a part of the NSF funded REU warm-water aquatic ecology site at Auburn University. The student led an independent research project focused on conservation genomics of the critically imperiled freshwater gastropod *Leptoxis compacta*, which was published in *PeerJ* with the student as lead author.

2018 - 2020 Conservation genomics of freshwater mussels

I advised a post-doctoral researcher, Nicole Garrison, on a variety of projects that focused on conservation genomics of freshwater mussels. This work had a large policy component as it included development of US Fish and Wildlife Service genetics management plans for threatened and endangered species.

2016 - 2018 Population genomics of Round Rocksnail.

I co-advised a graduate student, Breanna Siple, on her project on conservation and landscape genomics of a federally threatened freshwater snail.

2014 - 2015 Biogeography and population genetics of Antarctic *Thouarella Octocorals*.

I mentored an undergraduate researcher, Katelynn Webster, on DNA extraction, PCR, sequence analysis, and composing a scientific manuscript.

Invited Talks

14. Whelan, N.V. Research, education, and future opportunities at The Ohio State University Museum of Biological Diversity. *Great Lakes Science Bootcamp*. Online. July 2026.

13. Whelan, N.V. Enabling freshwater conservation with systematics, genetics, and partnerships. *Weekly Seminar Series*. Miami University. Oxford, Ohio. April 2026.

12. **Whelan, N.V.** Conservation genetics research is essential for freshwater mollusk conservation: recent applications and surprising data. **President's Symposium. American Malacological Society Annual Meeting.** Tuscaloosa, Alabama. August 2023.
11. **Whelan, N.V.** Systematics and conservation of freshwater gastropods. **Smithsonian Institution National Museum of Natural History, No Bones About It Seminar Series.** March 2021.
10. **Whelan, N.V.** Genomics as an essential tool for freshwater mussel conservation. **National Conservation Training Center Freshwater Mussel Webinar Series.** July 2020. Recorded at <https://training.fws.gov/topic/online-training/webinars/freshwater-mussel-conservation.html>
9. **Whelan, N.V.** Modeling substitutional heterogeneity and its impact on inferring relationships. **www.phyloseminar.org.** October 2016. Recorded at <https://www.youtube.com/watch?v=VLcn-iQq5CQ>
8. **Whelan, N.V.** Gastropod and ctenophore phylogenetics: difficult questions require fresh perspectives. **Academy of Natural Sciences Research Seminar Series.** Philadelphia, Pennsylvania. September 2016.
7. **Whelan, N.V.** Is everything we think we know about animal phylogeny and snail life history wrong? **Auburn University Fisheries and Aquaculture Seminar Series.** Auburn, Alabama. August 2016.
6. **Whelan, N.V.** (speaker), K.M. Kocot, A.B. Kohn, T.P. Tatiana, K. Mukherjee, P. Williams, C. Mills, G. Paulay, L.L. Moroz, K.M. Halanych. Body plan and lifestyle evolution of ctenophores. **Ctenopalooza.** St. Augustine, Florida. March 2016.
5. **Whelan, N.V.** Utilizing genomics to understand aquatic invertebrate evolution. **Field Museum, A. Watson Armour Seminar Series.** Chicago, Illinois. December 2015.
4. **Whelan, N.V.** (speaker), K.M. Kocot, K.M. Halanych. Resolving the metazoan tree of life with advanced bioinformatics pipelines and phylogenetic methods. *Society for Integrative and Comparative Biology Annual Meeting.* Origins of neurons and parallel evolution of nervous systems: the dawn of neuronal evolution symposium. West Palm Beach, Florida. January 2015.
3. Halanych, K.M. (speaker), K.M. Kocot, **N.V. Whelan.** Early animal relationships: alternative hypotheses and character inference. *Society for Integrative and Comparative Biology Annual Meeting.* Origins of neurons and parallel evolution of nervous systems: the dawn of neuronal evolution symposium. West Palm Beach, Florida. January 2015.
2. **Whelan, N.V.** Systematics and Life History Evolution of Pleurocerid Snails (Cerithioidea: Pleuroceridae). *Truman State University Biology Seminar Series.* Kirksville, Missouri. April 2014.
1. **Whelan, N.V.** Systematics and egg laying evolution of Pleuroceridae (Gastropoda: Cerithioidea). *Annual Meeting of the American Malacological Society.* Conchologists of America grant winners symposium. Philadelphia, Pennsylvania. June 2012.

Oral Presentations (*Undergraduate author, **Graduate student author, ‡postdoc author)

60. **Whelan, N.V.** Should the phylogenetic position of Ctenophora be controversial? *Ctenopalooa Ctwo.* Moss Landing, California. August 2026.

59. George, A.L., **N.V. Whelan (speaker)**, J.J. Apodoca, S. Chance, A.R. Drake, D.A. Foltz II, J. Garrido, J. Hall, K.A. Lamonthé, S.L. Lance. Development of an inventory and practitioner network to improve aquatic conservation translocations. *Freshwater Mollusk Conservation Society Biennial Symposium*. Ypsilanti, Michigan. May 2025.
58. Krause, A.L. (speaker)***, P.D. Johnson, M.L. Buntin, **N.V. Whelan**. Population genomics and parentage relationships between wild and captive cohorts of the federally endangered Alabama Lampmussel, *Lampsilis virescens*. *Freshwater Mollusk Conservation Society Biennial Symposium*. Ypsilanti, Michigan. May 2025.
57. Steed, A.E. (speaker)***, **N.V. Whelan**. Population genomics of the Tennessee River drainage endemic Smooth Rocksnail, *Leptoxis virgata* (Gastropoda: Cerithioidea: Pleuroceridae). *Freshwater Mollusk Conservation Society Biennial Symposium*. Ypsilanti, Michigan. May 2025.
56. Strasko, E.K. (speaker)***, J.T. Garner, P.D. Johnson, **N.V. Whelan**. Population structure of the endemic and federally threatened Painted Rocksnail, *Leptoxis coosaensis*. *Freshwater Mollusk Conservation Society Biennial Symposium*. Ypsilanti, Michigan. May 2025.
55. Donohoo, S.A. (speaker)***, J.S. Tiemann, **N.V. Whelan**. Systematics of the critically imperiled pleurocerid genus *Lithasia*. *Freshwater Mollusk Conservation Society Biennial Symposium*. Ypsilanti, Michigan. May 2025.
54. **Whelan, N.V. (speaker)**, A. Baldwin*, J.T. Garner, P.D. Johnson, E.E. Strong. Phylogenetics and museum records reveal misplaced conservation priorities for Pleuroceridae snails. *Society for Integrative and Comparative Biology Annual Meeting*. Atlanta, Georgia. January 2025.
53. Hannah, A.P. (speaker), J.W. Mays, C. Ruble, **N.V. Whelan**. Range-wide population structure and genomic diversity of the federally threatened Spotfin Chub (*Erimonax monachus*). *Southeastern Fishes Council Annual Meeting*. Little Rock, Arkansas. November 2024.
52. **Whelan, N.V. (speaker)**, A. Baldwin*, J.T. Garner, P.D. Johnson, E.E. Strong. The Unsolved Case of the Black Mudalia, *Elimia melanoides* (Gastropoda: Cerithioidea: Pleuroceridae). *American Malacological Society Annual Meeting*. Pasadena, California. August 2024.
51. Steed, A.E. (speaker)***, **N.V. Whelan**. Population genomics show high genetic diversity of *Leptoxis virgata* in a fragmented riverscape. *American Malacological Society Annual Meeting*. Pasadena, California. August 2024.
50. Donohoo, S.A. (speaker)***, J.S. Tiemann, **N.V. Whelan**. Systematics of the critically imperiled pleurocerid genus *Lithasia*. *American Malacological Society Annual Meeting* Pasadena, California. August 2024.
49. Rezac, C.R., R.J. Ellwanger, S.A. Donohoo**, P.D. Hartfield, A.S. Ruppel, M.D. Wagner, **N.V. Whelan (speaker)**. Surveys that prioritize many sites have better outcomes: a case on finding the Big Black Rocksnail. *Society for Integrative and Comparative Biology Annual Meeting*. Seattle, Washington. January 2024.
48. Donohoo, S.A. (speaker)***, P.D. Johnson, **N.V. Whelan**. Springs as islands: island biogeography and conservation genomics of two narrow-range endemic pleurocerid snails. *Society for Integrative and Comparative Biology Annual Meeting*. Seattle, Washington. January 2024.

47. Hannah, A.P. (speaker), J.W. Mays, C. Ruble, **N.V. Whelan**. Range-wide landscape genomics of the federally threatened spotfin chub (*Erimonax monachus*). *Society for Integrative and Comparative Biology Annual Meeting*. Seattle, Washington. January 2024.
46. Adcock, M.C. (speaker)** , S. Julian, M.L. Bartron, B.T. Watson, **N.V. Whelan**. Population genetic analysis of *Alasmidonta varicosa* (Brook Floater) in the U.S. and Canada. *Freshwater Mussel Recovery Implementation Workshop*. Moncton, New Brunswick, Canada. November 2023.
45. Hannah, A.P. (speaker), K. Inoue, D.J. Berg, K. Moles, T. Lane, **N.V. Whelan**. Molecular Ecology of the federally endangered freshwater mussel *Cumberlandia monodonta*. *Freshwater Mollusk Conservation Society Biennial Symposium*. Portland, Oregon. April 2023.
44. Adcock, M.C. (speaker)** , K.R. Moles, N.L. Garrison, S.A. Donohoo** , **N.V. Whelan**. Population genomics of the endangered freshwater mussel *Arcidens wheeleri*, in the Little River Arkansas. *Freshwater Mollusk Conservation Society Biennial Symposium*. Portland, Oregon. April 2023.
43. Donohoo, S.A. (speaker)** , P.D. Johnson, **N.V. Whelan**. Springs as islands: island biogeography and conservation genomics of two narrow-range endemic pleurocerid snails. *Freshwater Mollusk Conservation Society Biennial Symposium*. Portland, Oregon. April 2023.
42. **Whelan, N.V.** (speaker), A. Baldwin* , P.D. Johnson, J.T. Garner, E.E. Strong. Systematics, taxonomy, and molecular ecology of Black Mudalia, *Elimia melanoides* (Gastropoda: Pleuroceridae). *Freshwater Mollusk Conservation Society Biennial Symposium*. Portland, Oregon. April 2023.
41. **Whelan, N.V.** (speaker), A.P. Hannah, K. Inoue, D.J. Berg. Molecular ecology of the federally endangered freshwater mussel *Cumberlandia monodonta*. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2023.
40. **Whelan, N.V.** (speaker), E.E. Strong, N.S. Gladstone** , J. Mays. Using genomics and environmental niche modeling to test the validity of a narrow-range endemic snail. *Evolution*. Cleveland, Ohio. June 2022.
39. **Whelan, N.V.** (speaker), N.S. Gladstone** , A.S. Williams, J. Mays. Morphological divergence corresponds to genome-wide genetic variation in a terrestrial snail. *Society for Integrative and Comparative Biology Annual Symposium* +. Online. January 2022.
38. Gladstone, N.S.** (speaker), M.L. Niemiller, B. Hutchins, B. Schwartz, A. Czaja, M.E. Slay, **N.V. Whelan**. Groundwater snail biodiversity and conservation in the United States and Mexico. *National Cave and Karst Symposium*. San Marcos, Texas. November 2021.
37. Gladstone, N.S.** . (speaker), P.D. Johnson, **N.V. Whelan**. Evolution of interspecific egg-laying strategies in the freshwater gastropod family Pleuroceridae (Caenogastropoda: Cerithioidea). *Freshwater Mollusk Conservation Society Biennial Symposium*. Online. April 2021.
36. **Whelan, N.V.**, P.D. Johnson, J.T. Garner, N. Garrison, E.E. Strong (speaker). Insights into the biology, biogeography, conservation and systematics of Pleuroceridae (Gastropoda: Cerithioidea) within a new phylogenomic framework. *Freshwater Mollusk Conservation Society Biennial Symposium*. Online. April 2021.
35. **Whelan, N.V.** Phenotypic plasticity in freshwater gastropods influences shell shape less than you think. *Freshwater Mollusk Conservation Society Biennial Symposium*. Online. April 2021.

34. **Whelan, N.V.** (speaker), W. Daniel. Roles and activities of the gastropod status and distribution committee. *Freshwater Mollusk Conservation Society Biennial Symposium*. Online. April 2021.
33. Ksepka, S. P. ** (speaker), B. H. Hickson, **N. V. Whelan**, & S. A. Bullard. A new species of *Myxobolus* Bütschli, 1882 (Bivalvulida: Myxobolidae) infecting stratum spongiosum of the imperiled sicklefin redhorse, *Moxostoma* sp. (Cypriniformes: Catostomidae) from the Little Tennessee River, North Carolina, USA. *Auburn Student Research Symposium*. Auburn, Alabama. March 2021.
32. **Whelan, N.V.** (speaker). Is phenotypic plasticity a common driver of shell shape variation in freshwater gastropods? *Society for Integrative and Comparative Biology Annual Meeting*. Online. January 2021.
31. Gladstone, N.G. ** (speaker), P.D. Johnson, **N.V. Whelan**. Evolution of egg laying behavior in a critically imperiled freshwater gastropod family (Cerithioidea: Pleuroceridae). *Society for Integrative and Comparative Biology Annual Meeting*. Online. January 2021.
30. **Whelan, N.V.** (speaker), A.S. Williams, C. Redak**, A.A. Wright*, N.L. Garrison*, K.M. Halanych, P.D. Johnson, J.T. Garner. Migration patterns and fragmentation drive landscape genomic patterns in freshwater snails. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2020.
29. Strong, E.E. (speaker), **N.V. Whelan**, N.L. Garrison*, P.D. Johnson, J.T. Garner. Targeted sequencing and phylogenomics of the critically imperiled Pleuroceridae (Gastropoda: Cerithioidea). *World Congress of Malacology*. Pacific Grove, California. August 2019.
28. **Whelan, N.V.** (speaker), N.L. Garrison*, J.T. Garner, P.D. Johnson, E.E. Strong. Phylogenomics reveals relationships and biogeographic patterns of a critically imperiled group of freshwater snails (Cerithioidea: Pleuroceridae). *Evolution*. Providence, Rhode Island. June 2019.
27. **Whelan, N.V.** (speaker), P.D. Johnson, J.T. Garner, E.E. Strong. More is better: hundreds of nuclear genes improve understanding of Pleuroceridae (Gastropoda: Cerithioidea) relationships. *Freshwater Mollusk Conservation Society Biennial Symposium*. San Antonio, Texas. April 2019.
26. Garrison, N.L. (speaker)*, P.D. Johnson, **N.V. Whelan**. Conservation genomic assessment of wild and captive populations of the Louisiana Pearlshell, *Margaritifera hembeli* (Conrad), using RADSEQ. *Freshwater Mollusk Conservation Society Biennial Symposium*. San Antonio, Texas. April 2019.
25. Warren, M.B. ** (speaker), **N.V. Whelan**, S.A. Bullard. Systematics of two species of elasmobranch blood flukes (Digenea: Aporocotylidae) suggest both host-switching and cophyly. *Southeastern Society of Parasitologists*. Starkville, Mississippi. April 2018.
24. Warren, M.B. ** (speaker), **N.V. Whelan**, R. Yanong, S.A. Bullard. When something seemingly simple turns out to be not simple: Mermithid-infected eastern grass shrimp (*Palaemonetes paludosus*) (Decapoda: Palaemonidae). *43rd Annual Eastern Fish Health Workshop*. Chattanooga, Tennessee. April 2018

23. Warren, M.B.** (speaker), **N.V. Whelan**, S.A. Bullard. Host-switching and cophyly among blood parasites of early branching gnathostomes. *This is Research Student Symposium*. Auburn Alabama. March 2018
22. **Whelan, N.V.** (speaker), B.N. Siple** , M.P. Galaska** , B.S. Helms, P.D. Johnson, K.M. Halanych. Populations of *Leptoxis ampla*, a federally threatened snail species, are surprisingly distinct. *Society for Integrative and Comparative Biology*. San Francisco, California. January 2018.
21. **Whelan, N.V.** (speaker), A. Maloy, M.Curtis, M. Bartron. Mitochondrial genome sequencing and phylogenetic placement of the functionally extinct Alabama Sturgeon. *Southeastern Fishes Council Annual Meeting*. Chattanooga, Tennessee. November 2017.
20. Johnson, P.D (speaker), **N.V. Whelan**, J. Archambault. Identification of research priorities for North American freshwater gastropods. *Society for Freshwater Science Annual Meeting*. Raleigh, North Carolina. June 2017.
19. Abdelrahman, H.A. (speaker), M. Clay*, C. Figiel, **N.V. Whelan**, B.S. Helms, J. Stoeckel. Use of environmental DNA (eDNA) and the electron transport system (ETS) assay to assess current and future spread of invasive *Orconectes virilis* in the southeastern United States. *Society for Freshwater Science Annual Meeting*. Raleigh, North Carolina. June 2017.
18. **Whelan, N.V.** (speaker), P.D. Johnson, J.T. Garner, K.M. Halanych, B.S. Helms, E.S. Strong. Applying genomics to advance our understanding of freshwater mollusks. *Freshwater Mollusk Conservation Society Biennial Symposium*. Cleveland, Ohio. March 2017.
17. Strong, E.S. (speaker), **N.V. Whelan**. Filling the gaps: the importance of dense geographic sampling for assessing the diversity of western North American *Juga* (Gastropoda: Semisulcospiridae). *Freshwater Mollusk Conservation Society Biennial Symposium*. Cleveland, Ohio. March 2017.
16. **Whelan, N.V.** (speaker), K.M. Halanych. Who Let the CAT Out of the Bag? Handling substitutional heterogeneity with data partitioning results in more accurate phylogenies. *Evolution*. Austin, Texas. June 2016.
15. **Whelan, N.V.** (speaker), K.M. Kocot, A.B. Kohn, T.P. Tatiana, K. Mukherjee, P. Williams, C. Mills, G. Paulay, L.L. Moroz, K.M. Halanych. Phylogenomics resolves relationships among major Ctenophora lineages. *Society for Integrative and Comparative Biology*. Portland, Oregon. January 2016.
14. **Whelan, N.V.** (speaker), K.M. Halanych. Model choice and metazoan phylogenomics: model complexity does not ensure accurate phylogenetic hypotheses. *The Origins of Metazoa International Workshop*. Giens, France. October 2015.
13. Johnson, P.D. (speaker), A.E. Bogan, K.M. Brown, N.M Burkhead, J.R. Cordeiro, J.T. Garner, P.D. Hartfield, D.A.W. Lepitzki, G.R. Mackie, E. Pip, T.A. Tarpley, J.R. Tiemann, **N.V. Whelan**, E.E. Strong. Update to the conservation status of freshwater gastropods of Canada and the United States. *American Malacological Society Annual Meeting*. Pellston, Michigan. August 2015.
12. **Whelan, N.V.** (speaker), K.M. Kocot, L.L. Moroz, K.M. Halanych. Error, signal, and the placement of Ctenophora sister to all other animals. *Evolution*. Guarujá, Brazil. June 2015.

11. **Whelan, N.V.** (speaker), P.D. Johnson, E.E. Strong. Draft genome assembly of *Leptoxis ampla* (Pleuroceridae): a resource for conservation studies. *Freshwater Mollusk Conservation Society Biennial Symposium and Joint Meeting with the Upper Mississippi River Conservation Committee*. St. Charles, Missouri. March 2015
10. Johnson, P.D. (speaker), A.E. Bogan, K.M. Brown, N.M Burkhead, J.R. Cordeiro, J.T. Garner, P.D. Hartfield, D.A.W. Lepitzki, G.R. Mackie, E. Pip, T.A. Tarpley, J.R. Tiemann, **N.V. Whelan**, E.E. Strong. Update to the conservation status of freshwater gastropods of Canada and the United States. *Freshwater Mollusk Conservation Society Symposium and Joint Meeting with the Upper Mississippi River Conservation Committee*. St. Charles, Missouri. March 2015.
9. **Whelan, N.V.** Modeling life history evolution of a critically imperiled family of freshwater gastropods. *Evolution*. Snowbird, Utah. June 2013.
8. **Whelan, N.V.** (speaker), E.E. Strong, P.D. Johnson. Morphology, molecules, and taxonomy: the pleurocerid problem. *Freshwater Mollusk Conservation Society Symposium*. Guntersville, Alabama. March 2013.
7. Strong, E.E. (speaker), J.T. Garner, P.D. Johnson, **N.V. Whelan**. Divergent haplotypes and implications for phylogeny of the Pleuroceridae using mitochondrial markers. *Freshwater Mollusk Conservation Society Biennial Symposium*. Guntersville, Alabama. March 2013.
6. **Whelan, N.V.** Systematics of *Leptoxis* (Gastropoda: Pleuroceridae). *American Malacological Society Annual Meeting*. Pittsburgh, Pennsylvania. July 2011.
5. **Whelan, N.V.** (speaker), P.M. Harris, P.D. Johnson. Conservation and systematics of *Leptoxis* (Gastropoda: Pleuroceridae). *Freshwater Mollusk Conservation Society Biennial Symposium*. Louisville, Kentucky. April 2011.
4. **Whelan, N.V.** Life history evolution of *Leptoxis* (Gastropoda: Pleuroceridae). *Young Systematists Forum*. London, United Kingdom. December 2010.
3. **Whelan, N.V.** Life history of *Leptoxis* (Gastropoda: Pleuroceridae). *Joint Meeting of the American Malacological Society and the Western Society of Malacologists*. San Diego, California. July 2010.
2. **Whelan, N.V.** (speaker), B. Hartwig, T. Blasingame, D.R. DeCock, J.C. Gering. Advances in the statistical methodology of phylogenetic community ecology. *Association of Southeastern Biologists Annual Meeting*. Birmingham, Alabama. April 2009.
1. **Whelan, N.V.** Team katydid: the math-bio experience. *21st Annual Student Research Conference*. Truman State University, Kirksville, Missouri. April 2008.

Poster Presentations (*Undergraduate student author; **Graduate Student Author)

11. Baldwin, A*, P.D. Johnson, J.T. Garner, E.E. Strong, **N.V. Whelan**. Molecular Ecology, Conservation, and Taxonomy of the freshwater snail black Mudalia, *Elimia melanoides*. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2023.
10. Gonzalez, R.M.*, P.D. Johnson, M. Brizidene, **N.V. Whelan**. Genomic analyses reveal high genetic diversity in a highly imperiled freshwater mussel. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2023.

9. Roop, H, M. Hamel, P. Sakaris, **N.V. Whelan**, A.S. Williams, M. Bartron, C. Rees, J. Page. Where have all the snakehead gone? Two years of monitoring a wild population of northern snakehead (*Channa argus*) in Gwinnett County, Georgia. *Annual Meeting of the Georgia Chapter of the American Fisheries Society*. Jekyll Island, Georgia. February 2022.
8. Wright, A.D.*, A.S. Williams, N.L. Garrison, **N.V. Whelan**. Conservation Genetics of *Leptoxis compacta*. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2020.
7. Redak, C.***, A.S. Williams, K.M. Halanych, **N.V. Whelan**. Genetic diversity and gene flow of an endangered freshwater snail, *Pleurocera foremani*. *Society for Integrative and Comparative Biology Annual Meeting*. Austin, Texas. January 2020.
6. Williams, A.S., **N.V. Whelan**. Assessment of hatchery contribution and genetic diversity of American shad (*Alosa sapidissima*) in the Edisto River, South Carolina. *Southeastern Fishes Council Annual Meeting*. Knoxville, Tennessee. November 2019.
5. Townsend, J.P.***, M.G. Tassia**, A. Damian-Serrano, **N.V. Whelan**, K.M. Halanych, A.M. Sweeney. A colorful, deep sea ctenophore species from the northwest Atlantic Ocean. *Society for Integrative and Comparative Biology Annual Meeting*. San Francisco, California. January 2018.
4. Webster, K.J.*, **N.V. Whelan**, K.M. Halanych. A molecular investigation into the biodiversity and biogeography of Antarctic *Thouarella* (Cnidaria: Octocorallia: Primnoidae). *Society for Integrative and Comparative Biology Annual Meeting*. West Palm Beach, Florida. January 2015.
3. **Whelan, N.V.**, E.E. Strong. Extreme mitochondrial and nuclear phylogenetic discordance in Pleuroceridae (Gastropoda: Cerithioidea). *Evolution*. Snowbird, Utah. June 2013.
2. **Whelan, N.V.**, A. Geneva, D.L. Graf. What if anything is a Unionid? *Ceolatura* Conrad 1852 and the monophyly of the Unionidae (Mollusca: Bivalvia: Unionoida). *London Malacological Society's Malacology Forum*. London, United Kingdom. November 2010.
1. **Whelan, N.V.**, B. Hartwig, T. Blasingame, D.R. DeCock, J.C. Gering. Effects of phylogenetic tree topology and local and regional species richness on NRI and NTI distributions. *Annual Conference for the Society of Mathematical Biology*. San Jose, California. August 2007.

Professional Service

- 2026-present** **Chair**. Gastropod Names Subcommittee, Freshwater Mollusk Conservation Society
- 2023-present** **Associate Editor**. *Systematic Biology*.
- 2023** **Invited ad hoc Reviewer**. National Science Foundation. (2 proposals from 2 panels)
- 2022** **Invited Panel Reviewer**. National Science Foundation. Division of Environmental Biology.
- 2022** **Invited Panel Reviewer**. National Science Foundation. Graduate Research Fellowship Program.
- 2021** **Invited Proposal Reviewer**. Society of Systematic Biologists Annual Student Research Grants.
- 2021** **Session Moderator**. Freshwater Mollusk Conservation Society Symposium.

- 2021** **Invited Panel Reviewer.** National Science Foundation. Systematics and Biodiversity Science Cluster
- 2021** **Invited Panel Reviewer.** National Science Foundation. Graduate Research Fellowship Program.
- 2020-2025** **Faculty Senator, School of Fisheries, Aquaculture, and Aquatic Sciences.** Auburn University Faculty Senate.
- 2020** **Invited Proposal Reviewer.** Society of Systematic Biologists Annual Student Research Grants.
- 2018-2025** **Voting Member.** Gastropod Names Subcommittee, Freshwater Mollusk Conservation Society.
- 2017** **Invited Panel Reviewer.** National Science Foundation. Systematics and Biodiversity Science Cluster
- 2017** **Invited Expert Reviewer.** Red Wolf Recovery Committee, United States Fish and Wildlife Service.
- 2017-2025** **Invited Internal Reviewer for Species Status Assessments.** U.S. Fish and Wildlife Service.
- 2017-2018** **Invited Proposal Reviewer.** Society of Systematic Biologists Annual Student Research Grants.
- 2016-present** **Editorial Board Member.** *Systematic Biology*
- 2016-2025** **Lead Geneticist.** Tennessee River Lake Sturgeon Recovery Committee.
- 2016-2025** **Lead Geneticist.** Gulf Striped Bass Recovery Committee.
- 2016-2020** **Lead Geneticist.** Sicklefin Redhorse Candidate Conservation Agreement.
- 2016-2018** **Member.** Snail Darter Recover Committee.
- 2013-2021** **Chair.** Gastropod Distribution and Status Committee, Freshwater Mollusk Conservation Society.
- 2012-present** **Appointed Member.** Alabama State Gastropod Conservation Priority Committee.
- 2012-present** **Invited Reviewer.** *American Malacological Bulletin, Biological Invasions, Bulletin of the Society of Systematic Biologists, Conservation Genetics, Current Biology, Communications Biology, Ecology and Evolution, Freshwater Mollusk Biology and Conservation, Freshwater Science, Journal of Heredity, Heredity, Hydrobiologia, Invertebrate Systematics, Journal of Molluscan Studies, Molecular Biology and Evolution, Molecular Ecology, Molecular Ecology Resources, North American Journal of Fisheries Management, Northeastern Naturalist, PLoS One, Polar Biology, PeerJ, Proceedings of the Royal Society B, Scientific Reports, Southwestern Naturalist, Subterranean Biology, Systematic Biology, Toxicon, US Fish and Wildlife Service Mollusk Recovery Plans and Threat Assessments, Zoological Journal of the Linnean Society, Zoological Studies.*
- 2011-2014** **Appointed Member.** Endangered Species Committee, American Fisheries Society.
- 2011-2013** **Student Councilor at Large.** American Malacological Society Executive Council.

Awards and Honors

- 2025** **Dean's Grantsmanship Award.** Auburn University.
- 2022** **Selected for Advanced Leadership Development Program.** U.S. Fish and Wildlife Service.

- 2020 Science Achievement Award.** Smithsonian Institution National Museum of Natural History.
- 2018 Against All Odds Award for commitment to freshwater mollusk conservation in the Cahaba River.** Cahaba River Society.
- 2016 30 under 30 award for exceptional young professionals.** Pi Kappa Phi.
- 2014 Annual Award for Outstanding Initial Contribution to Malacology.** London Malacological Society.
- 2012 Ralph L. Chermock Prize for Most Outstanding Graduate Student.** The University of Alabama, Department of Biological Sciences.
- 2010 2nd Prize, Best Student Talk.** The Systematics Association.
- 2010 Best Student Oral Presentation.** American Malacological Society.
- 2009 Honorable Mention, Best Oral Presentation.** Association of Southeastern Biologists.
- 2008 Presidents Academic Achievement Award.** Truman State University.
- 2004 President's Combined Ability Scholarship.** Truman State University.
- 2004 Foreign Language Scholarship.** Truman State University.
- 2004 Missouri Bright Flight Scholarship.**
- 2004 National Eagle Scout Association Scholarship.**

Outreach

- 2026 Lead Organizer, Museum of Biological Diversity Open House.** Largest outreach event that The Ohio State University College of Arts and Sciences puts on.
- 2026 Podcast Interview, Voices of Excellence.** <https://podcast.osu.edu/voices-of-excellence/nathan-whelan-rediscovering-the-extinct>
- 2026 STEAM on the Quad at OSU-Lima.** Led hand-on activities geared towards educating kids about biodiversity.
- 2025 - present Lead and Give Tours at Museum of Biological Diversity.** Museum-wide or collections-specific tours to external groups.
- 2025 The Ohio State University WestFest.** Organized and ran an activity on freshwater mussel biology for over 450 attendees.
- 2021 Invited professional speaker for Tri-Beta honor society at Columbus State University.**
- 2021 Virtual lesson to 9th grade students at Babcock High School in Punta Gordo Florida.**
- 2019 Staff training for the Cahaba Environmental Center.** One day training on freshwater mollusk biodiversity and identification in the Cahaba River in central Alabama.
- 2019 - present Warm Springs National Fish Hatchery Kids-Go-Fish Day.** Annual program to educate kids about recreational fishing and teach basic fishing skills.
- 2019 Auburn University School of Fisheries, Aquaculture, and Aquatic Sciences Field Day.** Day-long exhibit on freshwater invertebrate ecology with live animal exhibits.
- 2016 - present Warm Springs National Fish Hatchery Open Houses.** Annual open house and educational event at Warm Springs National Fish Hatchery.
- 2016 Greater East Alabama Science and Engineering Fair Judge.** Judged science fair projects of middle school students from across eastern Alabama.

- 2015** **NPR's Science Friday video feature.**
I was interviewed for the weekly national broadcast. A video produced by Science Friday about my research was featured on their website:
<http://www.sciencefriday.com/video/08/03/2015/the-unlikely-tale-of-a-tenacious-snail.html>
- 2014 - 2016** **Icy Invertebrates Outreach Team.** Auburn University.
Made presentations to K-12 students in Alabama about Antarctic invertebrate biology.
- 2014 - 2016** **AU Explore.**
Annual science outreach program to middle school students in eastern Alabama.
- 2008 - 2012** **Volunteer Elementary School Science Tutor.** Tuscaloosa's One Place.

Selected Bioinformatics Repositories Available from <http://github.com/nathanwhelan>

- 10. 3RAD_scripts** Scripts for automatically processing raw 3RAD data with STACKS.
- 9. AHE-data-processing** Pipeline and associated scripts for analysis of anchored hybrid enrichment data
- 8. population_genomics_scripts** R scripts for population and landscape genomic analyses.
- 7. 2bRAD-procesing** Bash scripts for pre-processing of 2bRAD population genetic data.
- 6. Generate-Sequences** Python and bash scripts for simulating phylogenomic-like datasets using known trees and Indel-Seq-Gen.
- 5. Order-Genes-by-Evolutionary-Rate** Python script for ranking evolutionary rate of a set of genes using single-gene trees.
- 4. Post-HaMStR-Orthology-Script** Shell script for initial orthology determination using homologous amino acid sequences inferred by HaMStR
- 3. Split-Supermatrix-Into-Partitions** R script for splitting a sequence super matrix into individual genes; used primarily to aid analyses on published datasets that require alignments of each gene.
- 2. Make-gene-list** Bash script to make a partition list for programs like PartitionFinder and BaCoCa from many single gene alignments.
- 1. Automate-PAML-codeml** Python script that automates evolutionary rate analyses for hundreds or thousands of genes using the PAML package codeml.

Field Experience and Research Cruises

- 2017 - present** *Freshwater mussel sampling.* Since 2017 I have worked on population genomics projects for freshwater mussels, which has included fieldwork. I have experience sampling threatened and endangered freshwater mussel species, including *Margaritifera hembeli*, *Epioblasma brevidens*, and *Hamiota altilus*.
- 2008 - present** *Freshwater gastropod sampling.* Since 2008 I have done frequent sampling for population genetics and systematics studies on freshwater gastropods. I have also done full drainage qualitative surveys, in collaboration with the Alabama Department of Conservation and Natural Resources. Sampling has included threatened and endangered species (e.g., *Leptoxis plicata*, *Athearnia anthonyi*, *Lioplax cyclostomaformis*).
- 2016** *Research Vessel Sharp.* Chief Scientist: Alison Sweeny. Atlantic Ocean
Midwater sampling of invertebrates with an emphasis on ctenophores and mollusks.

2014

Research Vessel Oceanus. Chief Scientist: Craig R. Smith. Pacific Ocean
Recovery of deep-sea landers and processing of samples, particularly *Xylophaga* bivalves
and *Osedax* annelids.